

BEST award research progress report

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I. GENERAL SETTING

We are interested in jointly modeling multiple series with Markov Switching Multifractal (from here referred as MSM) model. For now Let's just consider the bivariate case with series α, β .

II. UNIVARIATE MODEL SPECIFICATION

The graph on the right hand side is the univariate specification of the MSM. First, we introduce the notation in the univariate case as a precursor to the bivariate specification.

M_t : latent volatility state for each frequency ($\bar{k} \times 1$)

$$M_t = (M_{1,t}, \dots, M_{\bar{k},t}) \in \mathbb{R}_+^{\bar{k}}$$

γ : Switching probability for each frequency ($\bar{k} \times 1$)

$$\gamma \equiv (\gamma_1, \gamma_2, \dots, \gamma_{\bar{k}}) \in (0, 1)^{\bar{k}}$$

$M(\theta)$: The distribution elements of M_t is drawn from

θ : Parameters of this distribution

$$m \sim M(\theta), E(m) = 1, \theta = \{m_0, 0.5\}$$

$$M(\theta) \equiv P(m, u) = \mathbb{1}_{\{u < 0.5\}} m_0 + \mathbb{1}_{\{u \geq 0.5\}} m_1$$

γ_k : Switching probability of frequency k

$$\gamma_k = 1 - (1 - \gamma_1)^{b^{k-1}}$$

$$\gamma_1 \in (0, 1), b \in (1, \infty)$$

$M_{k,t \rightarrow t+1}$: The dynamics of frequency k from t to $t+1$

$$P(M_{k,t+1}|M_{k,t}, \gamma_k) = M_{k,t} + \mathbb{1}_{u < \gamma_k} (m - M_{k,t})$$

$\sigma(M_t)$: latent volatility state at t

$$\sigma(M_t) = \bar{\sigma} \left(\prod_{i=1}^{\bar{k}} M_{i,t} \right)^{\frac{1}{2}}, \bar{\sigma} \in \mathbb{R}_+$$

r_t : returns at t (later referred to as Y_t)

$$r_t \sim N(0, (\sigma(M_t))^2)$$

Thus the univariate MSM has one model parameter $\bar{k}$ (= number of frequencies the volatility will be decomposed to), as well as four latent time homogenous parameters $\phi = \{\theta, \bar{\sigma}, \gamma_1, b\}$ and latent states M_t across time.

Sampling of the posterior from this model can be implemented as follows. If we define the information available up to time t as D_t we have

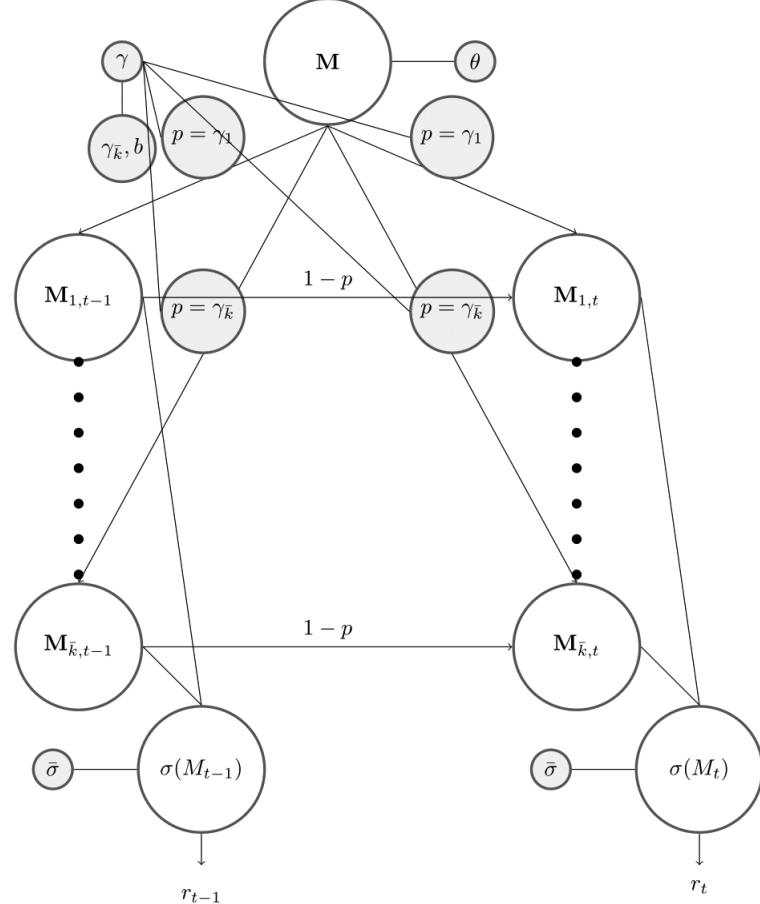


Fig. 1: DAG of MSM

$$\begin{aligned} P(M_t|D_t, \phi) &= P(M_{1,t}|D_t, \phi) \cdots P(M_{\bar{k},t}|D_t, \phi) \\ &\quad \because \text{independence among frequencies} \\ P(M_{t+1}|D_{t+1}, \phi) &\propto P(Y_{t+1}|M_{t+1}, \phi) P(M_{t+1}|D_t, \phi) \\ &\propto P(Y_{t+1}|M_{t+1}, \phi) P(M_{1,t+1}|D_t, \phi) \\ &\quad \cdots P(M_{\bar{k},t+1}|D_t, \phi) \\ &\propto P(Y_{t+1}|M_{t+1}, \phi) \sum_{M_{1,t}} P(M_{1,t+1}|M_{1,t}, \phi) \\ &\quad P(M_{1,t}|D_t, \phi) \cdots \sum_{M_{\bar{k},t}} P(M_{\bar{k},t+1}|M_{\bar{k},t}, \phi) \\ &\quad P(M_{\bar{k},t}|D_t, \phi) \end{aligned}$$

Thus we sample ϕ from appropriate priors and simulate posteriors $P(M_t|D_t, \phi)$ for $t = 1 : T$ and update ϕ with Metropolis-Hastings step. If necessary, backward sampling of

M_t 's can also be implemented in the recursive fashion similar to above.

III. OUR PROPOSED BIVARIATE EXTENSION

In order to define bivariate distribution $M_t = (M_t^\alpha, M_t^\beta)$, we introduce arrival parameter $1_{k,t}^\alpha, 1_{k,t}^\beta$ for $k = 1 : \bar{k}$. Then we assume that the dynamics of M_t follows the structural form below

$$\begin{aligned}
 1_{k,t}^\alpha | \gamma_k^\alpha &\sim \text{Bernoulli}(\gamma_k^\alpha) \\
 P(M_{k,t+1}^\alpha | M_{k,t}^\alpha, 1_{k,t}^\alpha) &= M_{k,t}^\alpha + 1_{k,t}^\alpha (m^\alpha - M_{k,t}^\alpha) \\
 \text{Then, e.g.} \\
 1_{k,t}^\beta | 1_{1:\bar{k},t}^\alpha, \gamma_k^\beta &\sim \begin{cases} \text{Bernoulli}(0.9), & (1_{1,t}^\alpha, 1_{2,t}^\alpha) = (1, 1). \\ \text{Bernoulli}(\gamma_k^\beta), & \text{otherwise.} \end{cases} \\
 P(M_{k,t+1}^\beta | M_{k,t}^\beta, 1_{k,t}^\beta) &= M_{k,t}^\beta + 1_{k,t}^\beta (m^\beta - M_{k,t}^\beta).
 \end{aligned}$$

where all $1_{k,t}^\beta$'s could potentially depend on some $1_{1:\bar{k},t}^\alpha$'s. This specification captures the local structure of the dependence between switches (or switch attempt since the state may still be the same even if $1_{k,t}^\alpha = 1$ as long as $M_{k,t}^\alpha = m^\alpha$). Now assume that we sampled M_t 's with the above structure and other relevant parameters to obtain $\sigma(M_t) = (\sigma(M_t^\alpha), \sigma(M_t^\beta))$ which is the std for each returns. In order to compute the joint likelihood, we need the covariance of these which can be represented as below:

$$\text{Cov}(r_t^\alpha, r_t^\beta) = \rho \sigma(M_t^\alpha) \sigma(M_t^\beta)$$

where ρ represents the correlation in the observational error which can either be set to zero or given another prior. This completes the specification of the joint distribution of returns.

Now we move on to the implementation of the sampling algorithm of this structural equations. Due to the necessity of sampling $1_{k,t}$'s, we no longer have exact form of $P(M_t | D_t)$'s since the marginal parts (like $\sum_{M_{1,t}} P(M_{1,t+1} | M_{1,t}, \phi) P(M_{1,t} | D_t, \phi)$) will now be computed using Monte Carlo sampling by sampling $1_{k,t}$'s. The full specification of the Gibbs sampling is as follows.

- 1) Sample $\phi = \{\theta^\alpha, \theta^\beta, \bar{\sigma}^\alpha, \bar{\sigma}^\beta, \gamma_1^\alpha, \gamma_2^\beta, b^\alpha, b^\beta, \rho\}$ from appropriate priors
- 2) For each univariate series α and β , MC sample $1_{k,t}$'s and $M_{k,t}$'s for all k, t .
- 3) Conduct a nonparametric regression of $1_{1:\bar{k},t}^\alpha$ on $1_{k,t}^\beta$ for all k by specifying an appropriate prior on this conditional distribution (cond-OPT?)
- 4) Depending on the captured conditional structure, resample $M_{k,t}^\beta$ and compute the joint likelihood of M_t .
- 5) Update ϕ with M-H step.
- 6) Back to 1 until good mixing.

IV. POTENTIAL ISSUES

We still have not paid any attention on issues such as identification problem etc. Also backward sampling part is still not considered.